MTH213 | MAT213: Elementary Differential Equations

Week 10: Second-Order Non-Homogeneous ODEs
Linear Independence, Wronskian, Technique 2: Variation of Parameters

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Introduction

What we have learned so far:

- Solving second-order homogeneous differential equations.
- Method of Undetermined Coefficients (Technique 1) for particular solutions.

Today we will learn:

- Linear dependence and independence of functions.
- Wronskian test for independence.
- Variation of Parameters (Technique 2) to find particular solutions.

Learning Objectives

By the end of this lecture, you should be able to:

- Explain linear dependence and independence of functions.
- Compute the Wronskian and use it to check independence.
- Apply Variation of Parameters to solve second-order non-homogeneous ODEs.

Why Linear Independence?

Suppose we have two solutions to a homogeneous equation: $y_1(x)$ and $y_2(x)$.

- Do they both provide "new" information about the solution?
- Or is one just a multiple of the other?

Key Idea

We need linearly independent solutions to form the general solution:

$$y_h = C_1 y_1 + C_2 y_2$$

Linear Dependence

Two functions y_1 and y_2 are **linearly dependent** if there exist constants c_1, c_2, not both zero, such that:

$$c_1 y_1(x) + c_2 y_2(x) = 0 \quad \text{for all } x \text{ in the interval.}$$

Interpretation: One function is a constant multiple of the other:

$$y_2 = k y_1$$

Linear Independence

Two functions are **linearly independent** if:

$$c_1 y_1(x) + c_2 y_2(x) = 0$$

implies that:

$$c_1 = 0 \quad \text{and} \quad c_2 = 0$$

Why it matters: Only linearly independent solutions give the full solution space for a second-order ODE.

Wronskian Definition

The Wronskian of two functions y_1, y_2 is:

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

Interpretation:

- $W(x) \neq 0$ at some point $\Rightarrow$ functions are linearly independent.
- $W(x) = 0$ everywhere $\Rightarrow$ functions are linearly dependent.

Example 1: Wronskian Calculation

Problem: Check if

$$y_1 = e^{2x} \quad \text{and} \quad y_2 = e^{-2x}$$

are linearly independent.

Step 1: Compute Derivatives

Compute the derivatives of the functions:

$$y_1' = \frac{d}{dx}(e^{2x}) = 2e^{2x}$$

$$y_2' = \frac{d}{dx}(e^{-2x}) = -2e^{-2x}$$

Step 2: Form the Wronskian

The Wronskian of two functions is defined as:

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

Substitute y_1, y_2, y_1', y_2':

$$W = \begin{vmatrix} e^{2x} & e^{-2x} \\ 2e^{2x} & -2e^{-2x} \end{vmatrix}$$

Step 3: Evaluate Determinant

$$W = (e^{2x})(-2e^{-2x}) - (2e^{2x})(e^{-2x})$$

$$W = -2 - 2 = -4$$

Observation: The Wronskian is *non-zero*.

Step 4: Conclusion

Since

$$W(y_1, y_2) \neq 0,$$

we conclude that:

$$\boxed{y_1 = e^{2x} \text{ and } y_2 = e^{-2x} \text{ are linearly independent.}}$$

Key Point

The Wronskian test is a reliable way to check independence of two solutions to a second-order ODE.

In-Class Activity: Linear Dependence

Task 1: Linear Dependence

Determine if the following sets of functions are linearly dependent.

1. $y_1 = x^2, \quad y_2 = 2x^2$
2. $y_1 = e^x, \quad y_2 = 3e^x$
3. $y_1 = x, \quad y_2 = x^2$
4. $y_1 = \sin x, \quad y_2 = 2\sin x$

In-Class Activity: Linear Independence

Task 2: Linear Independence

Check if the following pairs of functions are linearly independent using the Wronskian:

1. $y_1 = x, \quad y_2 = x^2$
2. $y_1 = e^x, \quad y_2 = xe^x$
3. $y_1 = \sin x, \quad y_2 = \cos x$

Hint: Compute the Wronskian

$$W(y_1, y_2) = y_1 y_2' - y_2 y_1'$$

If $W \neq 0$, the functions are independent.

Example 1: $y_1 = x, y_2 = x^2$

Step 1: Compute derivatives

$$y_1' = 1, \quad y_2' = 2x$$

Step 2: Compute Wronskian

$$W = y_1 y_2' - y_2 y_1' = x(2x) - x^2(1)$$

Step 3: Simplify

$$W = 2x^2 - x^2 = x^2$$

Conclusion: $W \neq 0$ for $x \neq 0$, so the functions are linearly independent.

Example 2: $y_1 = e^x, y_2 = xe^x$

Step 1: Compute derivatives

$$y_1' = e^x, \quad y_2' = e^x + xe^x$$

Step 2: Compute Wronskian

$$W = y_1 y_2' - y_2 y_1' = e^x(e^x + xe^x) - (xe^x)(e^x)$$

Step 3: Simplify

$$W = e^{2x} + xe^{2x} - xe^{2x} = e^{2x}$$

Conclusion: $W \neq 0$, so e^x and xe^x are linearly independent.

Example 3: $y_1 = \sin x, y_2 = \cos x$

Step 1: Compute derivatives

$$y_1' = \cos x, \quad y_2' = -\sin x$$

Step 2: Compute Wronskian

$$W = y_1 y_2' - y_2 y_1' = (\sin x)(-\sin x) - (\cos x)(\cos x)$$

Step 3: Simplify

$$W = -\sin^2 x - \cos^2 x = -1$$

Conclusion: $W \neq 0$, so $\sin x$ and $\cos x$ are linearly independent.

Review: Method of Undetermined Coefficients

- Suitable for ODEs of the form

$$y'' + py' + qy = f(x)$$

  where $f(x)$ is a polynomial, exponential, sine, cosine, or combination.
- Idea: Assume a particular solution y_p with unknown constants.
- Solve for constants by substituting y_p into the ODE.

Limitations of Undetermined Coefficients

- Works only when $f(x)$ is of standard forms:

$$e^{kx}, \quad \sin ax, \quad \cos bx, \quad x^n, \quad \text{or combinations}$$

- Fails for functions like:
  - $\ln x$
  - $\tan x$
  - e^{x^2}
  - $\sec kx$
  - $\arctan x$
- **Variation of Parameters works for all $f(x)$ where integration is possible.**

Why Variation of Parameters?

- Sometimes undetermined coefficients does **not** work.
- This happens when $f(x)$ is not a standard type.
- **Idea:** Instead of constants c_1, c_2 in y_h, we use functions $u_1(x), u_2(x)$.

Step 0: Our ODE

We consider a second-order linear ODE:

$$y'' + p(x)y' + q(x)y = f(x)$$

Homogeneous solution:

$$y_h = c_1 y_1(x) + c_2 y_2(x)$$

For Variation of Parameters, replace constants by functions:

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$$

Goal: find $u_1(x)$ and $u_2(x)$ so that y_p satisfies the ODE.

Step 1: Set Up the System

- Impose a simplifying condition:

$$u_1' y_1 + u_2' y_2 = 0$$

- Plug y_p into the ODE:

$$u_1' y_1' + u_2' y_2' = f(x)$$

- System of equations to solve:

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f(x) \end{cases}$$

Step 2: Solve for u_1', u_2'

- Using Cramer's rule:

$$u_1' = -\frac{y_2 f(x)}{W(y_1, y_2)}, \quad u_2' = \frac{y_1 f(x)}{W(y_1, y_2)}$$

- Step 3: Integrate to find functions:

$$u_1 = -\int \frac{y_2 f(x)}{W} dx, \quad u_2 = \int \frac{y_1 f(x)}{W} dx$$

- Step 4: Particular solution:

$$y_p = u_1 y_1 + u_2 y_2$$

- Step 5: General solution:

$$y = y_h + y_p = c_1 y_1 + c_2 y_2 + y_p$$